

Visualising GAP objects: current status and the future

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GAPDays Summer 2026

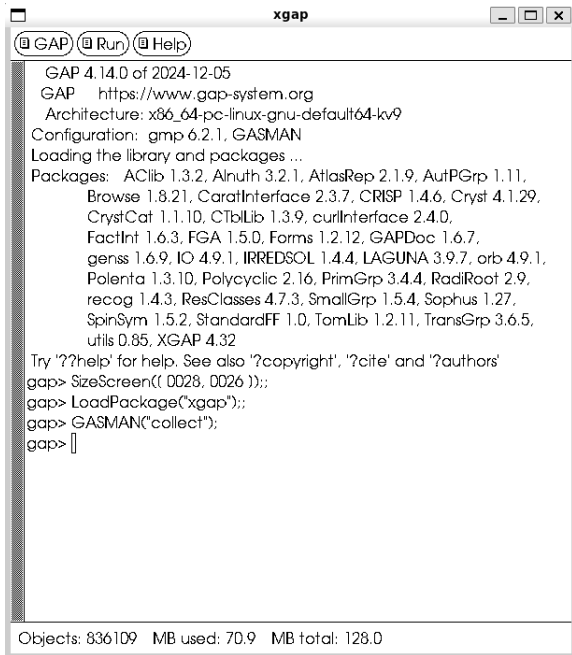
TODO: motivation

Currently there are multiple packages, which provide functionality to visualise GAP objects. First we will give an (possibly incomplete) overview of them with some examples.

Packages

Provides a GUI in which e.g. subgroup lattices can be investigated.
Needs to be started separately.

```
schnellecom@CURIE:/opt/gap/pkg/xcgap$ ./xcgap.sh
```

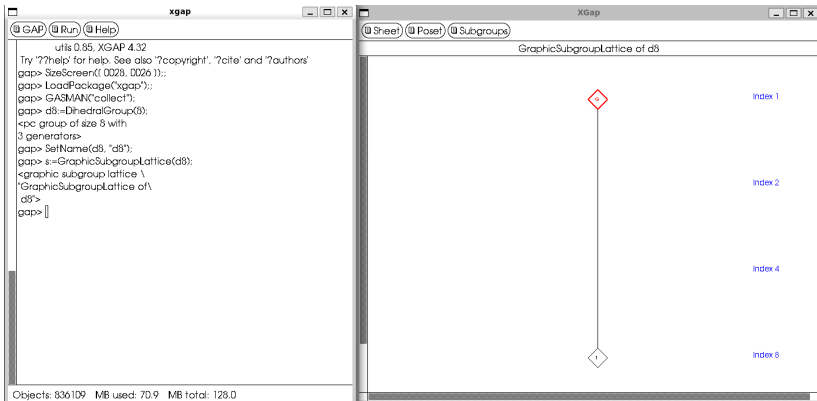


```
xgap

GAP 4.14.0 of 2024-12-05
GAP   https://www.gap-system.org
Architecture: x86_64-pc-linux-gnu-default64-kv9
Configuration: gmp 6.2.1, GASMAN
Loading the library and packages ...
Packages:  AClib 1.3.2, Alnuth 3.2.1, AtlasRep 2.1.9, AutPGrp 1.11,
           Browse 1.8.21, CaratInterface 2.3.7, CRISP 1.4.6, Cryst 4.1.29,
           CrystCat 1.1.10, CTblLib 1.3.9, curlInterface 2.4.0,
           FactInt 1.6.3, FGA 1.5.0, Forms 1.2.12, GAPDoc 1.6.7,
           genss 1.6.9, IO 4.9.1, IRREDSOL 1.4.4, LAGUNA 3.9.7, orb 4.9.1,
           Polenta 1.3.10, Polycyclic 2.16, PrimGrp 3.4.4, RadiRoot 2.9,
           recog 1.4.3, ResClasses 4.7.3, SmallGrp 1.5.4, Sophus 1.27,
           SpinSym 1.5.2, StandardFF 1.0, TomLib 1.2.11, TransGrp 3.6.5,
           utils 0.85, XGAP 4.32

Try '??help' for help. See also '?copyright', '?cite' and '?authors'
gap> SizeScreen(( 0028, 0026 ));
gap> LoadPackage("xgap");
gap> GASMAN("collect");
gap>

Objects: 836109  MB used: 70.9  MB total: 128.0
```



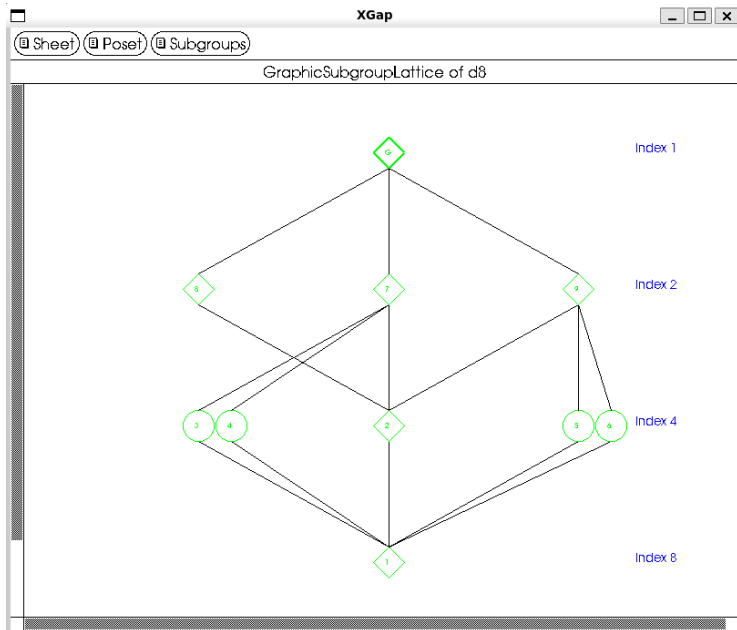
XGap

Sheet Poset Subgroups

All Subgroups
Centralizers
Centres
Closure
Closures
Commutator Subgroups
Conjugate Subgroups
Cores
DerivedSeries
DerivedSubgroups
Fitting Subgroups
Intermediate Subgroups
Intersection
Intersections
Normalizers
Normal Closures
Normal Subgroups
Sylow Subgroups
SelectedGroups to GAP
InsertVertices from GAP
Start Logging
Stop Logging

SubgroupLattice of d8

The diagram shows a subgroup lattice for the d8 group. It consists of two diamond-shaped nodes connected by a vertical line. The top node is red and contains the letter 'g', with 'Index 1' to its right. The bottom node is white and contains the number '1', with 'Index 8' to its right.



Advantages

- Direct GUI to use
- Freely movable visualisation

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Challenges

- Needs to be started standalone
- Needs to be adjusted to every OS

francy (by Manuel Martins)

Idea

Allow the XGAP system to run in a Jupyter Notebook.

Status

I was not able to get this running.

```

LoadPackage("digraphs"); LoadPackage("francy");

G := SymmetricGroup(3); as := AllSubgroups(G); nodes := [];
d := Digraph(as, {H, K} -> IsSubgroup(H, K));

vertices := DigraphVertices(d); edges := DigraphEdges(d);

canvas := Canvas(Concatenation("Subgroups Digraph of ",
    String(G)));
graph := Graph(GraphType.DIRECTED);
Add(canvas, graph);

customMessage := FrancyMessage(FrancyMessageType.INFO,
    "Simple Groups", "A group is simple if it is nontrivial
    and has no nontrivial normal subgroups.");

IsGroupSimple := function(i)
    Add(canvas, simpleGroupMessage);
    if IsSimpleGroup(as[i]) then
        Add(canvas, FrancyMessage("Simple",
            Concatenation("The vertex ", String(i),
                ", representing the subgroup ", String(as[i]),
                ", is simple.")));
    else
        Add(canvas, FrancyMessage("Not Simple",
            Concatenation("The vertex ", String(i),
                ", representing the subgroup ", String(as[i]),
                ", is not simple.")));
    fi;
    return Draw(canvas);
end;

```

```
for i in vertices do
  nodes[i] := Shape(ShapeType.CIRCLE, String(i));
  Add(nodes[i], Menu("Is this subgroup simple?",
    Callback(IsGroupSimple, [i])));
  Add(graph, nodes[i]);
od;

for i in edges do
  Add(graph, Link(nodes[i[1]], nodes[i[2]]));
od;

Draw(canvas);
```

(from [1])

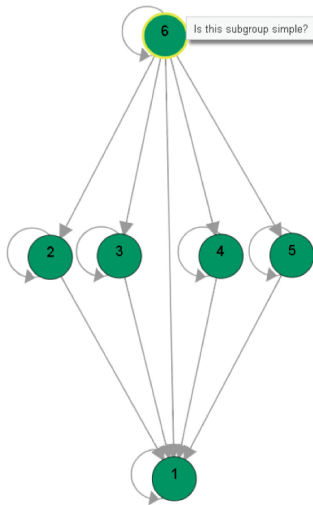
Francy

Subgroups Digraph of $\text{SymmetricGroup}([1..3])$

Simple Groups A group is simple if it is nontrivial and has no nontrivial normal subgroups.

Simple The vertex 2, representing the subgroup $\text{Group}([2,3])$, is simple.

Not Simple The vertex 6, representing the subgroup $\text{SymmetricGroup}([1..3])$, is not simple.



(from [1])

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- Runs in browser
- Highly customisable

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Other packages

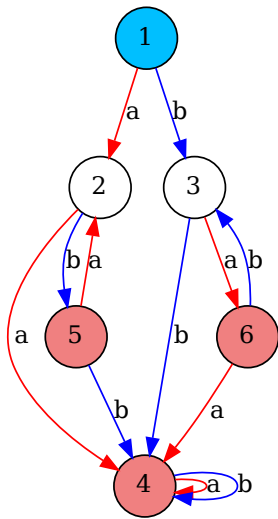
- jupyterviz
- semigroupviz

SgpViz (by Manuel Delgado and José João Morais)

Semigroup visualisation

- Can visualise e.g. Cayley graphs
- Using graphviz

```
gap> f := FreeMonoid("a","b");;
gap> a := GeneratorsOfMonoid( f )[ 1 ];;
gap> b := GeneratorsOfMonoid( f )[ 2 ];;
gap> r:=[a^3,a^2], [a^2*b,a^2], [b*a^2,a^2], [b^2,a^2], [a*b*a,a], [b*a*b,b] ;;
gap> b21:= f/r;
<fp monoid on the generators [ a, b ]>
gap> DrawCayleyGraph(b21);
```



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- Outputs PDFs

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Challenges

- Needs additional installation of graphviz on system

Automata (by Manuel Delgado, Steve Linton and José João Morais)

Automata visualisation

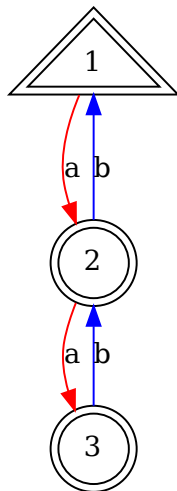
- Using graphviz

Automata (by Manuel Delgado, Steve Linton and José João Morais)

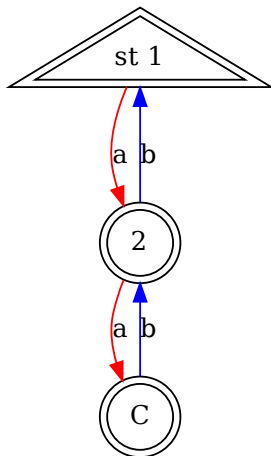
Automata visualisation

- Using graphviz

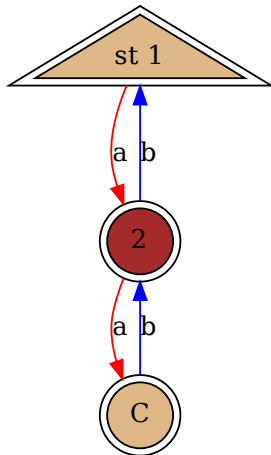
```
gap> x:=Automaton("det",3,2,[ [ 2, 3, 0 ], [ 0, 1, 2 ] ],[ 1 ],[ 1, 2, 3 ]);;
gap> DrawAutomaton(x);
```




```
gap> DrawAutomaton(x, ["st 1", "2", "C"]);
```



```
gap> DrawAutomaton(x,["st 1", "2", "C"],[[2],[1,3]]);
```



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graphviz (by James D. Mitchell and Matthew Pancer)

Digraphs visualisation

- (soon) extension to the Digraphs package
- Also uses graphviz standard

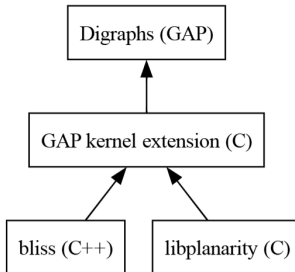
graphviz (by James D. Mitchell and Matthew Pancer)

Digraphs visualisation

- (soon) extension to the Digraphs package
- Also uses graphviz standard

```
gap> D := Digraph([], [1], [2], [2]);  
<immutable digraph with 4 vertices, 3 edges>  
gap> SetDigraphVertexLabels(D,  
> ["Digraphs (GAP)",  
> "GAP kernel extension (C)",  
> "bliss (C++)",  
> "libplanarity (C)"]);  
gap> gv := GraphvizVertexLabelledDigraph(D);  
<graphviz digraph "hgn" with 4 nodes and 3 edges>  
gap> GraphvizSetAttr(gv, "node [shape=box]");  
<graphviz digraph "hgn" with 4 nodes and 3 edges>  
gap> GraphvizSetAttr(gv, "rankdir", "BT");  
<graphviz digraph "hgn" with 4 nodes and 3 edges>
```

```
gap> Print(AsString(gv), "\n");  
//dot  
digraph hgn {  
  node [shape=box] rankdir=BT  
  1 [label="Digraphs (GAP)"]  
  2 [label="GAP kernel extension (C)"]  
  3 [label="bliss (C++)"]  
  4 [label="libplanarity (C)"]  
  2 -> 1  
  3 -> 2  
  4 -> 2  
}
```



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- Uses graphviz standard
- Is a package extension
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- Visualisations can be iterated

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Challenges

- Needs GAP function for everything in graphviz standard
- No immediate way to show results without additional installation on system

```
gap> Splash(gv);  
Error: /tmp/gaptempdirv6HuGo/hgn.dot: syntax error in line 4 near '('
```